

BEKSULTANOV ASANALI - 20232014 REPORT FOR PART B

Comparing the BLEU scores of the global model without input feeding and with input feeding we got this result.

Model Configuration	Input Feeding	BLEU Score	Validation Loss
Global Attention	No	0.2263	2.0757
Global Attention	Yes	0.2158	2.1308
Local Attention	Yes	0.1477	2.5320

It is mostly surprising because input feeding should improve the result, because input feeding is expected to improve performance by effectively increasing the model's capacity. By concatenating the previous time-step's context vector with the current token embedding, the decoder has explicit information about where it attended in previous steps. This additional "memory" helps the model make more informed predictions, reducing redundant repetitions and also providing much more context when generating each new token.

However I believe that here we have a relatively small dataset consisting only with 30000 examples, and given the fact that original input feeding paper was trained with 1 million sentences, I believe that our model has no time to optimize all its parameters.

Also in our dataset we have quite short sequences (as dataset shows maximum 205 symbols), so here we are losing the main benefit of input feeding, because the main problem benefit is coverage tracking for long sequences where model might repeat the translation of the same source words or miss some parts of the sentence. With short sequences that we have in our dataset, this problem is not severe, so we don't get much benefit from that.

Maybe increasing our epochs number or adjusting other hyperparameters will have a positive impact and the parameters will be able to train well.